

Hammad Farooq

Computational Scientist, Genomics and Bioinformatics

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Computational Scientist with expertise in statistical genetics, computational genomics, and machine learning. Experienced in large-scale bulk and single-cell multi-omics analysis, integrating ATAC-seq, RNA-seq, Hi-C, GWAS, and QTL data to interpret regulatory variation and identify target genes. Strong background in Python and R for scalable, reproducible workflows on high-performance computing (HPC) systems.

Education

University of Illinois Chicago

Expected June 2026

Ph.D. in Bioinformatics

Chicago, IL, USA

- **Thesis:** Predicting 3D chromatin structures to uncover roles of chromatin hubs in gene regulation.
- **Advisors:** Constantinos Chronis, Jie Liang

FAST National University of Computer and Emerging Sciences

2014-2018

Bachelor of Science in Computer Science

Islamabad, Pakistan

- **Final Year Project:** Predicting the pathogenicity of protein-coding mutations using Natural Language Processing.

Technical Skills

Single Cell Genomics: Single cell RNA seq, ATAC seq, and Hi C analysis including QC, clustering, cell type annotation, differential expression, and multimodal integration of expression, chromatin accessibility, and 3D genome data.

Machine Learning and AI for Regulatory Genomics: Design and interpretation of interpretable machine learning and deep learning models for enhancer promoter prediction, regulatory grammar inference, and noncoding variant impact analysis.

Computational Genomics and Precision Genetics: Large scale integration of functional genomics with GWAS and QTL data to interpret variant function, resolve LD structure, and identify disease relevant regulatory mechanisms.

Bioinformatics Frameworks: PLINK2, VCFtools, Seurat, Scanpy, Bioconductor, scikit learn, PyTorch, ChromBPNet, FiNeMo, MoDISco, ChromHMM, Juicer. Custom pipeline development for reproducible and scalable multi omics processing, regulatory annotation, and model evaluation in Python and R.

Population and Multi Omics Analysis: Population scale statistical genetics analyses using GWAS summary statistics, QTL mapping, variant set tests, and polygenic score modeling in large cohorts such as GTEx and All of Us.

High Performance Computing: Workflow automation with Parsl, containerized environments, GPU-based model training, and large-scale simulations and analysis of omics datasets on leadership-class systems (Aurora, Polaris).

Projects

Predicting Target Genes of Non Coding Regulatory Variants | Hi C, eQTL, GWAS, Causal Modeling

- Developed a multi omics framework integrating Hi C, eQTL, and GWAS data to prioritize causal noncoding regulatory variants and nominate enhancer linked target genes, with validation using CRISPR based perturbation datasets.

Variant Driven Regulatory Interaction Modeling [M39891345](#) | Hi C, eQTL, Single Cell Expression

- Developed a computational method integrating Hi C and eQTL data to infer variant associated regulatory interactions that modulate gene expression at single cell resolution, enabling mechanistic interpretation of regulatory effects.

Impact of LMNA Variant on Regulatory Architecture and AF Risk | ATAC seq, RNA seq, GWAS, Polygenic Scores

- Integrated RNA seq, ATAC seq, and variant level regulatory features with large scale biobank cohorts including All of Us to assess atrial fibrillation risk, combining rare variant driven chromatin effects with polygenic risk scores to identify disease relevant regulatory mechanisms.

Enhancer Gene Target Prediction | Python, R, Scikit Learn, ATAC seq, ChIP seq, Hi C

- Developed a machine learning model integrating chromatin accessibility, histone modification signals, and Hi C contacts to predict enhancer gene regulatory relationships across diverse cellular contexts.

Modeling Lamina Chromatin Interactions in Genome Architecture [M40027763](#) | Chromatin Organization

- Analyzed lamina associated chromatin features to define their role in large scale genome organization and transcriptional regulation.

Research & Professional Experience

Computational Biology University of Illinois Chicago

Aug 2019 – Present

Research Assistant

Chicago, IL, USA

- Led independent computational analyses on gene regulation and 3D chromatin architecture, with a focus on interpreting the functional impact of noncoding regulatory variants.
- Developed and maintained scalable analysis pipelines and worked closely with experimental collaborators to support genomics driven hypothesis generation and validation.

Computational Biology Research Lab at NUCES-FAST

Aug 2018 – Aug 2019

Research Assistant

Islamabad, Pakistan

- Contributed to computational biology and machine learning projects, applying deep learning to histopathological image analysis and protein structure prediction, and performing pharmacovigilance analyses using social media data.

Automotive Artificial Intelligence

Jan 2018 – Feb 2018

AI Developer Intern

Islamabad, Pakistan

- Designed and implemented components of an AI based intrusion detection system, applying machine learning methods to cybersecurity data.

Hackathons

ALCF INCITE GPU Hackathon

Argonne National Laboratory (Lemont, IL), May 2025

- Ported FoldRec C++ code from OpenACC to OpenMP offload for Intel GPUs on Aurora supercomputer.
- Integrated Parsl for scalable task management; optimized I/O and profiled performance using DAOS and unitrace.

ALCF INCITE GPU Hackathon

Argonne National Laboratory (Lemont, IL), May 2024

- Ported FoldRec C++ code from CPU to OpenACC offload for NVIDIA GPUs on Polaris & Theta supercomputer.
- Achieved 10x speedup by parallelizing key bottleneck loops and profiling with Nsight Systems and HPCToolkit.

Selected Publications

H. Farooq, L. Du, P. Delafrouz, W. Jiang, C. Chronis, J. Liang. "A New Approach for Discovering Functional Links Connecting Non-Coding Regulatory Variants to Gene Targets." *bioRxiv*, 2024.

L. Du, H. Farooq, P. Delafrouz, J. Liang. "Structural basis of differential gene expression at eQTLs loci from high-resolution ensemble models of 3D single-cell chromatin conformations." *Bioinformatics* 41(2): btaf050, 2025.

P. Delafrouz, H. Farooq, L. Du, A. Ma, J. Liang. "Effects of Lamina-Chromatin Attachment on Super Long-Range Chromatin Interactions." *bioRxiv*, 2025.

A. Owais, H. Farooq, H. Chen, P.K. Thami, et al. "Gene-gene interactions between a LMNA variant and common polymorphisms drive early-onset atrial fibrillation." *medRxiv*, 2025.

S. Priyadarshi, K.H. Bhat, E.B. Drake, S. Naiyer, H. Farooq, X. Qu, J.F. Cantillo, et al. "Igh nuclear speckle association promotes V (D) J recombination." *Nat Commun* (2026).

K.H. Bhat, S. Priyadarshi, S. Naiyer, X. Qu, H. Farooq, et al. "An Igh distal enhancer modulates antigen receptor diversity by determining locus conformation." *Nat Commun* 14(1): 1225, 2023.

H. Farooq, J.S. Niaz, S. Fakhar, H. Naveed. "Leveraging digital media data for pharmacovigilance." *AMIA Symp Proc*, 2020: 442.

N. Rehmat, H. Farooq*, S. Kumar, S. Hussain, H. Naveed. "Predicting the pathogenicity of protein-coding mutations using NLP." *IEEE EMBC*, pp. 5842–5846, 2020.

* Co-First Author

Honors & Awards

Travel Awards: CSHL Epigenetics & Chromatin (2024); BPS Annual Meeting (2023); UIC Graduate Student Council (2023, 2024)

Presentation Awards: 1st Place Oral – UIC CBQB Research Day (2022); 1st Place Poster – UI Cancer Center Symposium (2022)

Academic Recognition: Dean's Honor List (4x); Undergraduate Research Grant – NGIRI, Pakistan (2017–18)

Competitions: Winner – NaSCON'17 Inter-FAST CS Quiz; PCIT Intra-College Quiz

Selected Presentations

Biophysical Society (BPS) Annual Meeting: Oral and Poster – 2025 (Los Angeles); Poster – 2023 (San Diego), 2022 (San Francisco), 2021 (Virtual)

American Society of Human Genetics (ASHG): Poster – 2025 (Boston) , Poster – 2024 (Denver)

Cold Spring Harbor Laboratory (CSHL): Poster – Epigenetics & Chromatin (2024), Genome Organization & Nuclear Function (2022)

American Medical Informatics Association: Oral – Summit 2024 (Boston); Annual Symposium 2020 (Virtual)

UIC CBQB Research Day: Oral – 2024 (4th Annual), 2022 (2nd Annual), 2021 (1st Annual)

Other Conferences: Poster – UI Cancer Center Gary Kruh Symposium (2022, 2025); Oral – UIC BME Research Symposium (2025)

Peer Review Service

American Medical Informatics Association (AMIA): Annual Symposium (2022, 2025, 2026), Clinical Informatics Conference (2025), Amplify Informatics Conference (2025)

IEEE Engineering in Medicine and Biology Society (EMBC): Annual Conference (2021, 2025, 2026)